

Food Donation & NGO Connect Platform

Project topic proposal · Review 0

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Food is wasted while people go hungry



Surplus on one side, hunger on the other.

Restaurants, households and events discard edible food every day, while NGOs cannot find it in time to collect it.

1

Donors and NGOs work in silos

There is no shared, structured record of what food is available or what is needed nearby.

2

Manual coordination is slow

Phone calls and message groups get buried and do not scale across a city.

3

No tracking or accountability

Pickups are not followed end to end, so nobody can confirm that food reached people.



Today's approach and its gaps

CURRENT PRACTICE

- Phone calls and WhatsApp groups link donors to NGOs
- Basic listing apps let donors post surplus food
- Volunteers match requests from memory and contacts
- NGOs keep checking back for new donations

GAPS WE SEE

- ✗ No central, searchable record of donors and NGOs
- ✗ No matching by location, quantity or expiry
- ✗ No automatic alerts when new food is listed
- ✗ No pickup status tracking or audit trail
- ✗ No verification of donors and NGOs



One platform to list, match and deliver

A relational-database web platform where donors list surplus food and the system connects it to the nearest suitable NGO.



STEP 1

List surplus

Donors post food type, quantity, location and expiry time.



STEP 2

Match with SQL

Joins and proximity filters find NGOs by distance, quantity and expiry.



STEP 3

Notify NGOs

Database triggers alert matching NGOs as soon as a listing is added.



STEP 4

Track pickup

A volunteer collects the food and delivery status is logged to the end.



What we aim to achieve

1

Build one central relational record of donors, NGOs and food listings

2

Automate donor–NGO matching by location, quantity and expiry

3

Notify NGOs instantly using database triggers

4

Track every pickup from listing to delivery

5

Apply ER modelling, 3NF, joins and indexing in the design



EXPECTED OUTCOME

Faster food rescue

A working web system where surplus food can be listed, matched to a nearby NGO, picked up and marked delivered, with less delay, less wastage and a complete audit trail.



How we plan to build it

PROPOSED TECHNOLOGY (to be finalised in Review 1)



FRONTEND

React



BACKEND

Spring Boot / Django



DATABASE

MySQL / PostgreSQL



TOOLS

Git & GitHub, MySQL Workbench

PROJECT PLAN

Review 0

Topic, problem and idea

Review 1

Abstract, scope, tech stack, architecture

Next reviews

Design, build, test and demo

Thank You

TOPIC PROPOSED

**Food Donation & NGO Connect
Platform**

Team 18 | Review 0 | Section 10

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